



Inputs

Active Client Layer Updates

candidate updates
from active clients

$$\Delta W_k^{(l)}$$

RawScore from Stage 3

trust-weighted
permission score

$$RawScore_k$$



Control & Decision

Privacy Sensitivity

exposure risk
of layer 1

$$S_{\text{priv}}^{(l)}$$

Utility Sensitivity

contribution to
global direction

$$S_{\text{util}}^{(l)}$$

Security Deviation Sensitivity

deviation from
trusted reference

$$S_{\text{sec}}^{(l)}$$

risk fusion

Total Defense Requirement

weighted fusion of three signals

$$S_{\text{total}}^{(l)} = w_p S_{\text{priv}}^{(l)} + w_u S_{\text{util}}^{(l)} + w_s S_{\text{sec}}^{(l)}$$

threshold control

Layer Threshold

admission barrier
for this layer

$$\theta^{(l)}$$

radius control

Clip Radius

maximum safe
update magnitude

$$C^{(l)}$$



Constrained Execution

survivor filtering



Survivor Selection

keep only qualified clients

$$\Phi^{(l)} = \{k : RawScore_k \geq \theta^{(l)}\}$$

reweight survivors



Weight Re-normalization

redistribute weights over survivors

$$\sum \tilde{w}_k^{(l)} = 1$$

clip magnitude



Dynamic L2 Clipping

project updates into safe radius

$$\|\hat{\Delta} \hat{W}_k^{(l)}\|_2 \leq C^{(l)}$$

aggregate securely



Secure Layer Aggregation

aggregate filtered clipped survivors

$$\Delta W_{\text{global}}^{(l)} = \sum \tilde{w}_k^{(l)} \hat{\Delta} \hat{W}_k^{(l)}$$



Global Layer Update

applied to global model

$$\Delta W_{\text{global}}^{(l)}$$